Type Deletion and Scope-Local Symbols

Compiler Sessional Online

Problem Statement

Task 1

Create a command called `DT` **(Delete Type)**. This will take an argument - type. Example -

```
DT INT
```

This will delete all symbols (in the entire symbol table) with the given type - `INT`

Task 2

Create a command called `IS` which is used to insert a *scope-local symbol*. These are symbols that are only visible to their **own** container scope but **neither by their descendant scopes nor by their ancestor scopes**. For example, consider the following example -

```
1    int x;    ----> let's say x is a scope-local symbol
2    {
3        ----> x is invisible here
4    }
```

Here if `x` is a scope-local symbol then it won't be visible in line 3. Because line 3 is in a **descendant** scope of the scope-local variable.

Restrictions

- You **can** add any attribute to any existing classes.
- STL not allowed
- Must compile with `-fsanitize=address`

Examples

Sample Input

```
3
I x VAR
I y INT
IS s VAR
L s
S
I z INT
DT INT
P A
L s
Q
```

Sample Output

```
 ScopeTable# 1 created
Cmd 1: I x VAR
 Inserted in ScopeTable# 1 at position 1, 1
Cmd 2: I y INT
 Inserted in ScopeTable# 1 at position 2, 1
Cmd 3: IS s VAR
 Inserted in ScopeTable# 1 at position 2, 2
Cmd 4: L s
 's' found in ScopeTable# 1 at position 2, 2
Cmd 5: S
 ScopeTable# 2 created
Cmd 6: I z INT
 Inserted in ScopeTable# 2 at position 3, 1
Cmd 7: DT INT
 Deleted 'z' from ScopeTable# 2 at position 3, 1
 Deleted 'y' from ScopeTable# 1 at position 2, 1
Cmd 8: P A
 ScopeTable# 2
 1-->
 2-->
 3-->
  ScopeTable# 1
  1--> <x,VAR>
  2--> <s,VAR>
  3-->
Cmd 9: L s
 's' not found in any of the ScopeTables
Cmd 10: Q
 ScopeTable# 2 removed
 ScopeTable# 1 removed
```

Marks Distribution

Task	Marks
Type deletion	6
Scope-local symbols	4
Total	**10**